A fast, self-certifying Gaussian posterior for MAGI ODE parameter inference

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Abstract

MAGI [1] infers ODE parameters from noisy, partially observed trajectories by placing a Gaussian process prior on the state and conditioning on the ODE holding at a discretisation grid. Its posterior is high dimensional — the states outnumber the parameters by two orders of magnitude — so practitioners reach either for a long MCMC run or for the Laplace approximation $\mathcal{N}(x_\star, H^{-1})$ at the maximum a posteriori point. We show the latter is worse than it looks. On a FitzHugh–Nagumo benchmark it is mis-centred by a *full posterior standard deviation* on two of the three ODE parameters while its credible intervals have very nearly the right width, so nothing in the output signals the error.

We give a pipeline that removes it in about one second. Three observations drive it. First, with the observation noise fixed the MAGI log-density is *exactly* a sum of squares, so the mode is a nonlinear least-squares problem and Gauss–Newton reaches $\|\nabla \log p\| \approx 10^{-7}$ in 0.05 s where a first-order optimiser stalls at 10^3; the same structure yields the exact Hessian from one Jacobian assembly rather than d autodiff passes. Second, the residual error is almost entirely in the mean, and the standard third-order Laplace correction removes 90–99% of it. Third, a second and independently derived mean — the stationary point of the reverse-KL Gaussian objective, solved by a Newton iteration preconditioned on a screened low-dimensional subspace — supplies a *reference-free* error estimate: the two derivations can only agree closely if both are near the truth, and their disagreement relative to the size of the correction separates the regimes where the expansion is valid from those where it is not by more than thirty orders of magnitude.

We also give an exactness condition. If the vector field is affine in the state at fixed parameters — linear compartment models, linear pharmacokinetics, any constant-coefficient system, and much weaker than joint affineness — then $p(X \mid \theta)$ is exactly Gaussian, the 325-dimensional problem is exactly a 3-dimensional one, and no approximation is required. The condition is testable in two Hessian evaluations.

We report several negative results, including that Stein variational gradient descent does not merely converge slowly here: started *at* a correct answer it moves away from it, degrading energy distance by a factor of 45–80 while holding marginal standard deviations at 0.99 of the reference. Follow-up work reproduces that to three significant figures and locates the cause, which is not specific to this posterior: the collapse is dimensional, and the bandwidth rule the method ships with makes it a factor of $0.582K/\ln K$ worse than the rule it is usually credited with. A corrected configuration exists and brings the ensemble to the reference's own agreement floor—but it degrades the parameter estimates on two of three systems, which is what the exercise is for, so we report it as a diagnosis rather than a recommendation.

1 Introduction

Inferring the parameters $\theta \in \mathbb{R}^p$ of an ODE system $\dot{x} = f(x, \theta)$ from noisy, sparsely sampled observations is a classical inverse problem made awkward by the need to solve the ODE inside the likelihood. MAGI [1] avoids the solver: it places a Gaussian process prior on each state component, discretises time onto a grid $I = \{t_1, \ldots, t_n\}$, and conditions on the ODE residual being small at every grid point. The unknown becomes the concatenation of the parameters and the state trajectory,

$$x = (\theta, X) \in \mathbb{R}^d, \qquad d = p + nD,$$

with D the number of state components. On the FitzHugh–Nagumo benchmark used throughout this paper, $p = 3$, $n = 161$, $D = 2$ and $d = 325$: the trajectory outnumbers the parameters 107 to one.

This dimensionality is the practical difficulty. A well-tuned Hamiltonian sampler works but costs minutes, and — as we document in Section 8 — a configuration that converges at one data density can fail to converge at another without that being obvious. The cheap alternative is the Laplace approximation at the mode. Section 3 shows that on this problem the Laplace approximation is wrong in a specific and dangerous way: the covariance is essentially right, so the credible intervals have the right *width*, while the mean is off by a full posterior standard deviation on two of the three parameters. An analyst reading the output sees nothing amiss.

Contributions.

1. A characterisation of the Laplace error on MAGI posteriors: it lives in the mean, not the covariance, and we localise the residual covariance error to a handful of directions that cancel out of every marginal (Section 3).
2. A deterministic pipeline — Gauss–Newton mode, exact structural Hessian, third-order mean correction — costing about one second at $d = 325$ and reducing the largest parameter error by 2 to 10× at every setting tested (Sections 2–4).
3. A reference-free certificate and the demonstration that a naive version of it fails: the plain Gaussian-VI fixed point diverges, and restricting it to a screened subspace is what makes it both convergent and cheap (Section 5).
4. An exact discrepancy identity for the Gauss–Newton Gaussian, and an exactness condition on the vector field under which no approximation is needed (Section 6).
5. A methodological point that materially changed our own conclusions twice: in $d = 325$ the usual covariance metrics are dominated by the reference chain's own sampling error and are meaningless without a calibrated floor (Section 3.1); and an error averaged over all d coordinates is dominated by trajectory states and can rank estimators backwards relative to the parameters one actually wants (Section 4.5).

2 The MAGI posterior as nonlinear least squares

2.1 Setup

Write $\mu, \dot{\mu} \in \mathbb{R}^{n \times D}$ for the GP prior mean and its time derivative on the grid, $C_d^{-1}, K_d^{-1} \in \mathbb{R}^{n \times n}$ for the prior and conditional-derivative precisions of component d, m_d for the corresponding conditional-mean operator, τ for the observation mask, y for the observations, σ for the observation noise scale, and β^{-1} for the prior tempering factor. The MAGI log-density is

$$-2\log p(x) = \beta^{-1}\sum_d \|X_{\cdot d} - \mu_{\cdot d}\|^2_{C_d^{-1}} + \sum_d \frac{\|\tau_{\cdot d} \odot (X_{\cdot d} - y_{\cdot d})\|^2}{\sigma_d^2} + \beta^{-1}\sum_d \|r_d(x)\|^2_{K_d^{-1}} + c(\sigma), \quad (1)$$

with ODE residual $r_d(x) = f_d(X,\theta) - \dot{\mu}_{\cdot d} - m_d(X_{\cdot d} - \mu_{\cdot d})$.

2.2 Exact sum-of-squares structure

Proposition 1 (Least-squares form)**.** *With σ held fixed, (1) is exactly a sum of squares. Writing $L_c L_c^\top = C^{-1}$ and $L_k L_k^\top = K^{-1}$ and $b = \beta^{-1}$,*

$$-2\log p(x) = \|R(x)\|^2 + const, \qquad R(x) = \begin{bmatrix} \sqrt{b}\, L_c^\top (X-\mu) \\ (X-y)/\sigma \quad \text{on observed entries} \\ \sqrt{b}\, L_k^\top \big(f(X,\theta) - \dot{\mu} - m(X-\mu)\big) \end{bmatrix} \in \mathbb{R}^{3nD}. \quad (2)$$

We verified (2) numerically against (1) to 8.4×10^{-12}. Two consequences carry the rest of the paper.

Only the third block is nonlinear, and only through f. Hence the residual Jacobian is assembled from n small *pointwise* derivatives $\partial f/\partial(X_i,\theta) \in \mathbb{R}^{D\times(D+p)}$ obtained by autodiff on the user's vector field, plus constant precomputed blocks. The Gauss–Newton and observation blocks contribute a constant term to $J^\top J$, so the normal equations are formed without ever materialising $J \in \mathbb{R}^{3nD\times d}$.

The second derivative of the residual is the second derivative of the ODE. Differentiating (2) twice,

$$\nabla^2 R_a = \sqrt{b}\,(L_k^\top)_a \,\nabla^2 f \qquad \text{(pointwise in the grid index)}, \quad (3)$$

so the exact Hessian of $U = -\log p$ is

$$H(x) = \nabla^2 U = J(x)^\top J(x) + \textstyle\sum_a R_a(x)\,\nabla^2 R_a(x), \quad (4)$$

in which the second term is n small $(D{+}p) \times (D{+}p)$ blocks scattered into place. This costs one Jacobian assembly rather than the d forward passes `jax.hessian` would take; we measured agreement with `jax.hessian` to 2.5×10^{-9}.

2.3 The Gauss–Newton mode

Proposition 1 makes the mode a nonlinear least-squares problem, so Gauss–Newton applies, damped in the manner of Levenberg and Marquardt, and converges quadratically near the solution. This matters more than it may appear. The FitzHugh–Nagumo Hessian is conditioned around 2×10^4, where a first-order optimiser stalls far from the mode *while appearing to converge*: Prodigy terminates at $\|\nabla \log p\| = 1.3 \times 10^3$, a log-density 8.7 nats below the mode, whereas Gauss–Newton reaches 2.5×10^{-7} in 0.05 s. Everything downstream — the metric, the correction, the certificate — inherits that error. An earlier iteration of this work was conducted on the stalled mode and reached materially different conclusions; the displacement from mode to posterior mean was inflated threefold.

Unknown σ is handled by block coordinate descent: Gauss–Newton on (θ, X) at fixed σ, alternating with the exact conditional maximiser $\sigma_d^2 = \sum_{\text{obs}}(X-y)^2/N_d$. Both blocks decrease the same objective. The whole iteration runs inside one `lax.while_loop` with no host synchronisation and one compilation per solver instance.

quantity	$\mathcal{N}(x_\star, H^{-1})$	+ third order	gold half (floor)
θ_b mean error	$-\mathbf{0.971}$ sd	-0.010 sd	$+0.017$ sd
θ_c mean error	$+\mathbf{1.032}$ sd	$+0.093$ sd	-0.008 sd
θ_a mean error	$+0.246$ sd	$+0.034$ sd	$+0.009$ sd
trajectory max mean error	1.015 sd	0.115 sd	0.042 sd
θ sd ratios	1.00/1.01/0.94	0.99/1.01/0.94	1.00/0.99/1.01
trajectory sd ratio (median)	0.983	0.985	0.993

Table 1: The Laplace error at the baseline data density. The interval *widths* are within a few percent of a second independent reference run; two of the three parameter estimates are centred a full standard deviation away.

3 Diagnosis: the error is in the mean

3.1 Calibrating the metrics

Before any comparison we must know what agreement is achievable. In $d = 325$ the affine-invariant covariance metrics are dominated by the error in the *reference's own* covariance estimate: with n_{eff} effective draws the eigenvalues of a sample covariance spread by roughly $\pm 2\sqrt{d/n_{\text{eff}}}$, and a metric that is an rms over all 325 of them inherits that.

We use three: the mean error bias $= \|\Sigma_g^{-1/2}(\mu_q - \mu_g)\|/\sqrt{d}$ in posterior standard deviations per dimension; the Förstner distance $\|\log(\Sigma_g^{-1/2}\Sigma_q\Sigma_g^{-1/2})\|_F/\sqrt{d}$; and $\mathrm{KL}(q \,\|\, \mathcal{N}(\mu_g, \Sigma_g))$. Splitting an eight-chain, 64,000-draw gold standard four-and-four gives Förstner 0.1538; inverting that against synthetic draws puts the chain's effective sample size at 27,791 (43%), from which the floor for a *noiseless* approximation scored against the full chain is Förstner 0.0767, bias 0.0042, KL 0.48. For scale, inflating the gold covariance uniformly by 1.25× scores Förstner 0.2231 — indistinguishable from what the Laplace covariance scores. Comparisons reported without this calibration are not interpretable, and we found this changed the sign of our reading of one experiment.

3.2 What the Laplace approximation gets wrong

Table 1 scores $\mathcal{N}(x_\star, H^{-1})$ against the gold standard on the quantities a practitioner reads.

This is the failure mode that matters in practice, because it is invisible. A user inspecting posterior summaries sees plausible intervals of plausible width.

3.3 Where the residual covariance error lives

Three independent lines say the covariance is not worth chasing.

The residual error is real but hides in the null space of every marginal. Scoring the Laplace covariance against each half of the gold chain separately gives per-direction log-variance errors correlated at 0.975 (a pure-noise control gives 0.852), so the error is genuine. But it concentrates in about twenty directions holding 17% of the posterior variance: binning by variance share, the top five directions (78.5% of the variance) have rms log-ratio 0.040, the next twenty (17.3%) have 0.245, and the remaining three hundred (4.2%) have 0.03. Those twenty are combinations that cancel out of every parameter marginal and every pointwise trajectory band, which is why Table 1 shows standard deviations within 1–6% while the aggregate Förstner distance sits at three times its floor.

At this data density, a method that demonstrably fixes marginal variances does not move the joint covariance. We built Tierney–Kadane Laplace marginals [2] along screened directions (Section 4.3) using a bordered Gauss–Newton profile. For a unit direction v the marginal of $z = v^\top(x-\mu)$ satisfies, to second order,

$$\log p_{\mathrm{marg}}(z) = -U_{\mathrm{prof}}(z) - \tfrac{1}{2}\log\det H_\perp(z) + \mathrm{const}, \qquad U_{\mathrm{prof}}(z) = \min\{U(x) : v^\top(x-\mu) = z\},$$

and two identities make it cheap: the constrained Gauss–Newton step is the unconstrained step plus a multiple of $A^{-1}v$, reusing the *same* Cholesky factor, and for unit v, $\det(N^\top H N) = \det(H)\,(v^\top H^{-1} v)$ for any orthonormal basis N of $v^\perp$, so the restricted determinant never requires the $(d-1)$-dimensional basis. At 0.31 s per direction it cut the mean marginal-variance error from 4.68% to 1.66%, fixing the two worst directions from +8.7% to +0.8% and +6.7% to +1.0%. Its effect on the joint covariance was Förstner $0.2243 \to 0.2237$, for 13 s. We do not recommend it *here*; Section 4.6 revises this for harder settings, where the Förstner distance turns out to be the wrong thing to have judged it on.

Sampled metrics cannot resolve the effect. At 800 draws the standard error of a variance-weighted ratio is comparable to the 3–5% effect being measured; it reported the correction as a slight loss while the analytic comparison showed each corrected direction moving decisively toward truth. Analytic approximations should be scored analytically.

4 Correcting the mean

4.1 Third-order Laplace expansion

The standard third-order correction to the posterior mean [2, 3] is

$$\mathbb{E}[x] \;\approx\; \mu_3 \;:=\; x_\star - \tfrac{1}{2}H^{-1}\nabla\,\mathrm{tr}\big(H^{-1}\nabla^2 U\big), \tag{5}$$

evaluated here with the exact Hessian (4). It costs one gradient of a scalar built from a Hessian — 0.39 s at $d = 325$ once compiled.

That (5) should be needed at all is worth making concrete. Along the softest eigendirection of H the potential is not remotely quadratic: its relative departure from the quadratic model is 1.85 at one posterior standard deviation, 19.7 at four and 114 at sixteen, while along stiff directions it is 0 to four decimals. That direction carries roughly 20% of the total posterior variance.

4.2 Gaussian variational inference, and why the obvious iteration diverges

Minimising $\mathrm{KL}(q\|p)$ over Gaussians gives, by the Bonnet and Price theorems,

$$\mathbb{E}_q[\nabla \log p] = 0, \qquad \Sigma^{-1} = \mathbb{E}_q[-\nabla^2 \log p], \tag{6}$$

so the Laplace approximation is precisely the zeroth iterate of a fixed point built from quantities already available. The natural iteration $\mu \leftarrow \mu + \Sigma\,\mathbb{E}_q[\nabla \log p]$ **diverges**, for two reasons with one cause.

The mean step overshoots by about 1.6× (measured step 0.2751 against a correct ≈ 0.17, with Monte Carlo error only 0.0025) because the correct Newton preconditioner is $\mathbb{E}_q[H]$, not $H(x_\star)$. These differ enormously: on the screened subspace we measure $\mathbb{E}_q[H]/H(x_\star)$ between 6 and 10 at the baseline setting and 2121 at the sparsest. The average curvature of the potential over the posterior is an order of magnitude above its curvature at the mode.

The covariance step explodes because a 325×325 matrix average over a few hundred samples has $O(\sqrt{d/n})$ eigenvalue noise that inversion amplifies; one step took the trace ratio from 1.007 to 2.125.

Remark 1 (Antithetic pairing is load-bearing). Our first estimator of $\mathbb{E}_q[\nabla \log p]$ produced noise four times the size of the signal. The $O(\delta)$ term dominates the variance and is odd, so evaluating at $\mu \pm \delta$ cancels it exactly. Spherical cubature is worse than useless here: its $2d$ nodes sit at $\sqrt{d} \approx 18$ posterior standard deviations along a single axis, deep in the region where the degree-six polynomial dominates, and it diverged immediately.

4.3 A curvature screen

Both failures are cured by refusing to estimate what the samples cannot support. Along eigenvector v_j of H, with unit equal to one posterior standard deviation $s_j = \lambda_j^{-1/2}$, define

$$q_j \;=\; \frac{1}{4} \sum_{t \in \{-2,-1,1,2\}} \left| \frac{U(\mu + t s_j v_j)}{t^2/2} - 1 \right|, \tag{7}$$

which is zero if and only if the slice is exactly quadratic. It costs *four batched log-density evaluations for all d directions* (0.15 s), ranks them at Spearman 0.760 against the true error, and its top five directions capture 71% of the variance-weighted Laplace error (top ten, 85%).

The screen is a good detector and a poor estimator: along the softest direction the slice is 185% steeper than quadratic at one standard deviation while the true marginal is only 8% too narrow, the gap being the complement relaxing as one moves. We use it only to choose a subspace.

4.4 Low-rank Gaussian VI

Let S index the m largest q_j and V_S the corresponding eigenvectors. Restricted to S, $\mathbb{E}_q[H]$ is an $m \times m$ block — a well-determined estimate instead of an impossible one — and because S is spanned by eigenvectors of H the corrected curvature is block diagonal in H's eigenbasis and inverts in closed form:

$$A = H + V_S \Delta V_S^\top, \qquad A^{-1} = \sum_{i \notin S} \frac{v_i v_i^\top}{\lambda_i} + V_S \big(\mathrm{diag}(\lambda_S) + \Delta\big)^{-1} V_S^\top. \tag{8}$$

Only $V_S^\top H(x) V_S$ is ever needed, so m Hessian-*vector* products per sample replace a $d \times d$ assembly. The iteration $\mu \leftarrow \mu + A^{-1}\mathbb{E}_q[\nabla \log p]$ then converges in three steps, the step size falling $0.166 \to 0.0147 \to 0.0034 \to 0.0004$.

4.5 Which mean to report

We obtain two corrected means: μ_3 from (5) and μ_{VI} from (8). Both are $O(\Lambda)$-accurate with different $O(\Lambda^2)$ residuals, so their midpoint is a natural third candidate.

Scored as an error averaged over all 325 coordinates the midpoint wins (0.0234 against 0.0268 at baseline). Scored on the ODE parameters the ranking reverses, and by a wider margin: see Table 2. The average is dominated by the 322 trajectory states; the three parameters are what the method exists to estimate. **We default to μ_3**, giving up 13% on the average to gain a factor of three on the parameters.

	largest $|\theta$ error$|$ (reference sd)			all-coordinate
setting	$x_\star$	μ_3	midpoint	bias, μ_3 vs mid
baseline	1.033	**0.099**	0.297	0.0268 vs **0.0234**
half	1.579	**0.164**	0.616	0.0934 vs **0.0508**
noisy	2.607	1.337	**1.069**	0.4370 vs **0.1719**
quarter	1.760	**0.446**	diverged	0.3356 vs —

Table 2: The two metrics rank the estimators oppositely. We report μ_3; the midpoint is available for callers whose target is the trajectory.

4.6 When the covariance does matter

The preceding subsection is a statement about the baseline data density, and it does not generalise. From the identity (9) the leading mean shift is driven by the cubic term and is $O(\Lambda)$, whereas a covariance correction requires the quartic term or the square of the cubic, both $O(\Lambda^2)$. The covariance error is therefore negligible at small Λ and grows *faster* than the mean error — so the two must converge as a problem becomes more non-Gaussian, and the question is only where the crossover lies.

Across our four settings, which span a factor of 200 in measured non-Gaussianity, the ratio of (mean error / its floor) to (covariance error / its floor) falls from 15.4 at baseline to 3.3–6.6 at the harder settings, as predicted. In the units a user reads, the largest error in a parameter standard deviation is

κ_S	baseline 10.4	half 45.8	noisy 166.8	quarter diverged
largest θ sd error, H^{-1}	6.9%	17.7%	**70.0**%	36.6%
after the profile correction	**0.9**%	**6.4**%	**22.0**%	**11.6**%
reference half-vs-half floor	1.8%	1.7%	3.6%	3.1%

At $\sigma = 0.5$ the Laplace standard deviation for θ_b is 1.70 times the reference's — a credible interval 70% too wide, against a half-vs-half ratio of 1.0035, so this is real. The covariance is emphatically worth chasing there.

The tool that fixes it is the profile construction dismissed above, applied along the parameter coordinate axes rather than along eigenvectors, with two repairs: the quadrature grid must be bracketed on the *profile's* own scale rather than the Laplace scale (at $\sigma = 0.5$ a ± 4.5 Laplace-sd grid reaches ± 7.6 true standard deviations, where $H_\perp$ loses positive definiteness and the Cholesky fails), and nodes at which $H_\perp$ is indefinite must be dropped rather than poisoning the direction. It costs about 1.8 s per parameter and takes θ_b from 1.700 to 0.960 at the noisy setting and from 1.366 to 1.033 at the sparsest.

It is not uniformly safe: at the noisy setting it degraded θ_c from 1.8% to 22% error while repairing θ_b from 70% to 4%. It improves the worst case at every setting we tried and should be run when κ_S is large, but it is a repair for a Gaussian that is already in trouble, not a routine step.

Our original dismissal of the method was an artefact of scoring it on the affine-invariant Förstner distance, which weights all 325 directions equally and is dominated by its own noise floor. The parameter marginals are what a practitioner reads, and on those the method works.

setting	reference-free				needs a reference		
	$q_{\max}$	κ_S	τ_{end}	ρ	$\lvert\theta\rvert$ err $x_\star$	$\lvert\theta\rvert$ err μ_3	$\widehat{R}$
baseline (41 obs)	4.95	10.4	$1{\cdot}10^{-4}$	0.198	1.033	0.099	1.030
half (21 obs)	20.8	45.8	$3{\cdot}10^{-4}$	0.278	1.579	0.164	1.121
noisy (σ 0.5)	48.9	167	$3{\cdot}10^{-2}$	0.244	2.607	1.337	1.025
quarter (11 obs)	932	$9.6{\cdot}10^{32}$	$5{\cdot}10^{34}$	$3.2{\cdot}10^{34}$	1.760	0.446	1.105

Table 3: Certificates against truth. The gate statistic ρ separates the regimes by more than thirty orders of magnitude, but what it diagnoses is the breakdown of the VI solve, not the validity of μ_3.

5 Certification without a reference

5.1 Certificates

Everything above needs a reference chain to score. For deployment the question is whether the method knows when it is failing. Four quantities, none requiring a reference:

- d_A: $\|H_{XX}(\theta, X_1) - H_{XX}(\theta, X_2)\|/\|H_{XX}\|$ at two states sharing one θ; zero exactly under Condition 1 below. Two Hessian evaluations.
- $q_{\max}$: the largest screen value (7). Four batched log-density evaluations.
- κ_S: $\max_j \mathbb{E}_q[H]_{jj}/H_{jj}$ over the screened subspace.
- ρ: the *gate statistic* $\|\mu_3 - \mu_{\text{VI}}\|_H/\|\mu_3 - x_\star\|_H$.

ρ is the important one. μ_3 is a Taylor truncation of the posterior mean; μ_{VI} is the exact stationary point of the reverse-KL Gaussian objective. Neither is the truth and they come from genuinely different arguments, so they can only agree closely if both are near it. Normalising by the size of the correction makes the statistic dimensionless and self-scaling.

5.2 What the gate does and does not diagnose

Table 3 gives the certificates and the errors they are meant to predict.

At the three denser settings ρ lies in 0.20–0.28; at the sparsest the VI iteration diverges and ρ exceeds 10^{34}. Any threshold in that gap works; we use 0.5.

We initially used ρ to *suppress* the correction, reporting $x_\star$ when the gate fired. Better reference chains showed this to be wrong. At `quarter`, μ_3 improves the largest parameter error from 1.760 to 0.446 even though the VI has diverged; suppressing costs 1.3 standard deviations of parameter accuracy to save about a fifth of a standard deviation averaged over 322 trajectory coordinates. The correction is therefore applied unconditionally and the gate *warns*: it reports that the low-rank VI has broken down and, with it, that the Gaussian over the trajectory block should not be trusted, and that the user should escalate to a sampler.

Remark 2 (A trust region destroys the statistic). Bounding the VI step keeps a diverging iterate near μ_3, which makes the two estimates look as though they agree: at `quarter` a trust region pulled ρ from 3×10^{34} down to 0.30, below the threshold and wrong. The runaway magnitude is the signal and must not be damped.

6 Theory

6.1 An exact discrepancy identity

Let $\delta = x - x_\star$ and let $E(\delta) = R(x_\star + \delta) - R_\star - J\delta$ be the residual's nonlinear remainder, with $R_\star = R(x_\star)$.

Proposition 2 (Exact discrepancy)**.** *Let $q = \mathcal{N}(x_\star, (J^\top J)^{-1})$ be the Gauss–Newton Gaussian. Stationarity at the mode gives $R_\star^\top J = 0$, whence, with no truncation of any kind,*

$$\log q(x) - \log p(x) \;=\; (R_\star + J\delta)^\top E(\delta) \;+\; \tfrac{1}{2}\|E(\delta)\|^2 \;+\; \textit{const.} \tag{9}$$

Proof. $-2\log p = \|R_\star + J\delta + E\|^2 = \|R_\star\|^2 + 2R_\star^\top J\delta + \delta^\top J^\top J\delta + 2(R_\star + J\delta)^\top E + \|E\|^2$. The cross term $R_\star^\top J\delta$ vanishes by stationarity, and $-2\log q = \delta^\top J^\top J\delta + \text{const.}$ □

By (3) the entire departure from Gaussianity is thus carried by $\nabla^2 f$.

6.2 An exactness condition

Condition 1 (Affine in the state)**.** $f(\cdot, \theta)$ is affine in the state X for every θ.

Proposition 3 (Exact conditional Gaussianity)**.** *Under Condition 1 the residual is affine in X at fixed θ, $R = R_0(\theta) + A(\theta)X$, so*

$$U(\theta, X) = U\big(\theta, X^\star(\theta)\big) + \tfrac{1}{2}(X - X^\star)^\top A^\top A(X - X^\star), \qquad X^\star(\theta) = -(A^\top A)^{-1}A^\top R_0,$$

is exactly quadratic in X. Hence $p(X \mid \theta)$ is exactly Gaussian with covariance $(A^\top A)^{-1}$, and integrating X out is exact:

$$\log p(\theta) = -U\big(\theta, X^\star(\theta)\big) - \tfrac{1}{2}\log\det\big(A(\theta)^\top A(\theta)\big) + \textit{const.} \tag{10}$$

The d-dimensional problem is then exactly a p-dimensional one, solvable by quadrature with no MCMC and no Gaussian assumption on θ; the only error is quadrature error, controlled by adding nodes until the answer stops moving.

Condition 1 is much weaker than requiring f affine in (X, θ) jointly: θ may still multiply states, as it does in our test case through $c(V + R)$. It covers linear compartment models, linear pharmacokinetics and every constant-coefficient system. Setting the FitzHugh–Nagumo cubic coefficient to zero satisfies it, and we measure $d_A = 0.00 \times 10^0$ exactly there against 1.30×10^{-1} for the true cubic field. Evaluating (10) with three structurally different quadrature rules — Gauss–Hermite 9^3, Gauss–Hermite 13^3 and a uniform 21^3 grid over ± 5 standard deviations — agrees to all six printed decimals (means $0.085448, 0.562092, 0.811322$), confirming convergence rather than a shared blind spot. The Laplace parameter means there are off by 0.008, 0.035 and 0.027 standard deviations.

6.3 Why we do not report an a priori bound

Identity (9) combined with $\|E(\delta)\| \leq \frac{1}{2}\Lambda\|\delta\|^2$, $\Lambda = \sqrt{b}\,\|L_k\|\,L_2$ and $L_2 = \sup\|\nabla^2 f\|$, gives a rigorous bound. It is not useful. Sweeping the cubic coefficient α from 0 to 1 we measure $\Lambda = 2012, 1588, 2548, 4173$ while the correction size varies as $0.005, 0.010, 0.111, 0.174$; the ratio $\|\mu_3 - x_\star\|/\Lambda$ varies by a factor of 17. A supremum over the posterior bulk is far too loose in 325

Algorithm 1: Deterministic MAGI posterior, $\approx 1.1\,\mathrm{s}$ at $d = 325$

Input: MAGI model; subspace size $m = 12$; antithetic pairs $N = 256$; gate 0.5
Output: $\mathcal{N}(\mu, \Sigma)$ and certificates

1 $x_\star \leftarrow$ Gauss–Newton on $\|R\|^2$ // 0.05 s
2 $H \leftarrow J^\top J + \sum_a R_a \nabla^2 R_a$ at $x_\star$; eigendecompose; $\Sigma \leftarrow H^{-1}$ // 0.30 s
3 $\mu_3 \leftarrow x_\star - \frac{1}{2}\Sigma\nabla\,\mathrm{tr}(\Sigma\,\nabla^2 U)$ // 0.39 s
4 $q_j \leftarrow$ slice-curvature screen (7) for all d directions // 0.15 s
5 $S \leftarrow$ indices of the m largest q_j
6 $\mu \leftarrow x_\star$
7 **for** $k = 1$ **to** 6 **do**
8 draw antithetic $\{\pm\delta_i\}_{i \le N}$ from $\mathcal{N}(0, \Sigma)$
9 $g \leftarrow$ mean of $\nabla \log p$ over the $2N$ points
10 $B \leftarrow$ mean of $V_S^\top H V_S$ by HVP; floor its spectrum
11 $\mu \leftarrow \mu + A^{-1} g$ with A^{-1} from (8) // no trust region
12 **end**
13 $\rho \leftarrow \|\mu_3 - \mu\|_H / \|\mu_3 - x_\star\|_H$; warn if $\rho \ge 0.5$
14 **return** $\mathcal{N}(\mu_3, \Sigma)$

dimensions. The measured certificates of Section 5 are the honest substitute: they are computed rather than bounded, and they were validated against references.

The same sweep does show the diagnostics are coherent and monotone: at $\alpha = 0$, $d_A = 0$, $\kappa_S = 1.18$ and $\rho = 0.0012$ — every certificate independently and correctly reporting that Laplace is exact — rising monotonically to $d_A = 0.130$, $\kappa_S = 10.4$, $\rho = 0.198$ at $\alpha = 1$.

7 The pipeline

Algorithm 1 assembles the pieces. Lines 1–3 are the deterministic answer and account for two thirds of the cost; lines 4–13 exist to certify it, and produce a second mean as a by-product. Nothing in the loop requires a $d \times d$ Hessian: line 10 needs only m Hessian-vector products per sample, and line 11 inverts an $m \times m$ block through (8). The covariance is H^{-1} throughout — Section 3 found nothing cheap that improves on it.

Three choices in the algorithm are less arbitrary than they look, and each was arrived at by a failure. The antithetic draw on line 8 is not variance reduction for its own sake but the difference between a convergent and a divergent iteration. The spectral floor on line 10 is what makes the Newton step a descent direction when sampling noise pushes an ill-conditioned block indefinite, which happens in single precision. And the absence of a trust region on line 11 is deliberate: bounding the step would make a diverging iterate resemble a converged one and destroy the statistic computed on line 13.

8 Experiments

8.1 Setup

FitzHugh–Nagumo, $\dot{V} = c(V - V^3/3 + R)$, $\dot{R} = -(V - a + bR)/c$, on $t \in [0, 20]$ with a 161-point grid, $d = 325$. Four settings vary the data: *baseline* (41 observations per component, $\sigma = 0.2$), *half*

(21), *quarter* (11), *noisy* (41, $\sigma = 0.5$). At `quarter`, 22 data points constrain 325 unknowns plus three parameters — close to unidentified, and not a regime in which any Gaussian approximation should be expected to work.

8.2 References, and a warning about them

References are NUTS [4] via BlackJAX, preconditioned by the exact Hessian: we sample $y = H^{1/2}(x - x_\star)$ with an identity mass matrix. The metric affects only efficiency, never the stationary distribution, so using the object under test as the preconditioner is not circular. The baseline reference is 400 chains of 150 draws ($\widehat{R} = 1.03$); the three harder settings use 64 chains of 2000, for reasons given below.

Credibility rests on the baseline, where an independent eight-chain gold standard exists: the preconditioned run matches it *better than the gold chain matches itself*, at bias 0.0056 (half-vs-half floor 0.0095), Förstner 0.1358 (floor 0.1538) and KL 1.50 (floor 1.90).

We stress a negative finding about the fallback. Our first references used 400 chains of 150 draws — the same total cost — and reached $\widehat{R}$ of 1.16, 1.48 and 1.77 on the three harder settings while giving 1.03 at baseline. Being the exact method does not make a particular run correct, and a configuration validated at one data density can silently fail at another. The longer configuration reaches $\widehat{R} = 1.03$, 1.12, 1.03, 1.11; `half` nonetheless reports 6.2% divergences, so we treat it as indicative. Conclusions in Table 3 that depend on `half` and `quarter` should be read accordingly.

8.3 Main result

Table 2 is the headline: the largest parameter error falls by factors of 10.4, 9.6, 1.9 and 3.9 across the four settings. In energy distance over the whole vector, measured in Mahalanobis coordinates against a floor set by scoring one 2000-draw subsample of the reference against another, the corrected Gaussian reaches 0.0397 at baseline against a floor of 0.0331 — at 2000 draws it is not distinguishable from the reference chain.

8.4 Ablations

Hyperparameters. $m = 12$ and $m = 24$ give *identical* results, so the screened subspace saturates — an independent confirmation that the screen finds the right subspace — while $m = 6$ is materially worse ($\rho = 0.064$ against 0.034). N from 256 to 2048 and three seeds move ρ by $\pm 3\%$. The knobs were held fixed across every setting reported, so no result here is the product of per-setting tuning.

Precision. Single precision is the production default and is essentially free: the correction agrees with its float64 value to 6.7×10^{-4} relative, a gap of 0.00007 posterior standard deviations against a correction of 0.17 and a floor of 0.006, and the resulting bias is identical to four decimals. The float32 mode stalls at $\|\nabla \log p\| = 2.9 \times 10^{-2}$ rather than 5.4×10^{-7}, far below the level that affects anything downstream. This does depend on forcing full-precision matmuls: the einsums in the log-density are matrix–vector contractions in the forward pass, but their VJPs are matrix–matrix products, so on tensor-core hardware the *gradient* loses about four significant digits by default ($5.1 \times 10^{-7} \to 3.9 \times 10^{-3}$) while the forward value is bit-identical. The value is not the thing to test. At the `noisy` setting the VI solve does diverge in float32 where float64 converges; the gate catches it.

8.5 Cost

Algorithm 1 costs about 1.1 s at $d = 325$, against 144 s for the shortest preconditioned NUTS run that is trustworthy at baseline and considerably more elsewhere. A cheaper NUTS configuration (400 chains, 50 warmup, 5 draws, 159 gradient evaluations, 21 s) already reaches the sampling floor in energy distance, so the honest comparison is roughly 20×, not 130×, if only distributional summaries are wanted — but that configuration provides no usable $\widehat{R}$.

9 Negative results

We record these because each cost us time and none is obvious in advance.

Stein variational gradient descent does not converge slowly here; it converges to the wrong thing. Starting mSVGD *at* the corrected Gaussian — a known-good answer — and running it, energy distance degrades from 0.085 to 3.80 (density-reweighted kernel [6]) or 6.89 (standard RBF), and the Stein-identity ratio falls from 1 to 0.24 and 0.019. The reweighted kernel is the more instructive failure: it holds marginal standard deviations at 0.991 of the reference — visually perfect credible intervals — while being far from the target. Since the starting point was correct, the motion is unambiguously away from it, and no initialisation strategy can help.

We have since reproduced that to three significant figures across a change of integrator and a change of GP hyperparameter fit, and the explanation we gave for it was wrong. It is not a property of the MAGI posterior: SVGD started at exact draws from a plain $\mathcal{N}(0, I_{325})$ collapses identically. The equilibrium variance obeys $\mathrm{Var}_{\mathrm{SVGD}}/\mathrm{Var} = \ln K/d$, because the median heuristic measures the bandwidth *from the ensemble*, so contraction tightens the bandwidth, which permits more contraction. Ba et al. [5] prove the $\Theta(1/d)$ scaling for the plain median heuristic, with constant $(e-1)^{-1}K/d$; the extra $1/\ln K$ that Liu and Wang recommend, and that our implementation used, costs a further $0.582K/\ln K$ — a factor of 39 at $K = 400$. Deleting it is one line, and improves the energy distance on these posteriors by 1.8–2.4×.

A configuration that works, and does not solve the problem we have. Running SVGD in coordinates whitened by H^{-1} at the mode, at a *fixed* bandwidth and a small fixed step, has a genuine attractor—reached from starts a factor of four too narrow and a factor of four too wide—and turns the shape error into a pure scale error. A scale error is one number, and the Stein ratio measures it without a reference, so rescaling the ensemble by $1/\sqrt{R}$ completes the correction: energy distance 0.57 to 0.96 times the K-particle floor on the three systems with usable references, against 1.03 to 1.65 for the Gaussian draws it started from. On HIV it beats 400 exact draws on a Kolmogorov–Smirnov comparison.

It is still not a recommendation, and the reason is worth stating. The largest *parameter* error goes from 0.077 to 0.242 on FitzHugh–Nagumo and 0.071 to 0.401 on Lorenz, and the rescaling does not repair it, because it corrects a scale and this is a location error. Only HIV improves ($0.096 \to 0.067$). The ensemble metric improves because it is dominated by the 322 trajectory states; the three numbers the exercise exists to estimate get worse. That is the same trade this paper declines in Section 4.5, and it is declined here for the same reason.

The Tierney–Kadane profile machinery is not worth it at the baseline density — correct, 13 s, no gain on the joint covariance — but this does not extend to harder settings, where it is the only thing we found that repairs a badly wrong parameter interval (Section 4.6). We record it here mainly because judging it on the Förstner distance was a mistake.

Unrestricted Gaussian VI diverges, and a trust region that stops it diverging destroys the diagnostic (Section 5).

Spherical cubature is unusable in this geometry: its nodes sit 18 posterior standard deviations out.

"Self-consistent Laplace" — recomputing H at μ_3 and reapplying (5) — diverges by roughly nine orders of magnitude, because (5) is derived *at the mode*, where $\nabla U = 0$; restoring the missing $-\Sigma\nabla U$ term simply recovers the VI iteration.

10 Limitations

The evidence is one ODE system at four data densities. The gate threshold sits in a gap spanning thirty orders of magnitude and is not delicate, but the claim that μ_3 improves the parameters *unconditionally* rests on four settings and should be re-examined on other vector fields, particularly stiff and multimodal ones. Two of the four reference chains are imperfect ($\widehat{R} = 1.12$ with 6.2% divergences, and $\widehat{R} = 1.11$).

The pipeline leaves the covariance at H^{-1}, which Section 4.6 shows is defensible only while κ_S is small. We do not currently fold the profile repair into the pipeline, because it is not uniformly safe and we have four settings on one vector field on which to judge it; a user whose κ_S is large should either run it per parameter and check, or use a sampler.

Unknown observation noise is handled in the mode but the Gaussian approximation over σ is weaker than over (θ, X), since σ is positivity-constrained and enters the log-density through a clip; the structural Hessian (4) does not cover it and we fall back to a generic one.

Finally, the pipeline returns a Gaussian. Where the posterior is genuinely non-Gaussian — which the certificates will say — there is no substitute for a sampler.

11 Conclusion

The MAGI posterior is exactly a nonlinear least-squares problem, and taking that structure seriously pays three times over: a mode accurate to 10^{-7} in 0.05 s, an exact Hessian from one Jacobian assembly, and an explicit characterisation of what makes the posterior non-Gaussian. The consequence of practical importance is that the default Laplace approximation is mis-centred by a full posterior standard deviation on the parameters of interest while looking entirely healthy, and that about a second of arithmetic fixes it and reports how far it can be trusted.

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